Inflation Without an Inflaton: Observer-Screen Synchronization in Observer-Patch Holography Cosmology

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Abstract

We formulate observer-screen synchronization as a proposed alternative to an inflaton in Observer-Patch Holography (OPH). The conditional source theorem fixes a screen amplitude from a primitive collar release law and a spectral tilt from an edge-center generator. Physical source dilation and radial cross-covariance tomography give two sufficient routes to a unique three-dimensional spectrum. Vanishing spatial curvature holonomy identifies a flat FLRW branch. On the sourced causal order the expansion history is a layer-dependent record density: the comoving read law generates the same finite order for every scale-factor profile, and the redshift between two epochs is the fourth root of a record-count ratio. These results specify how a finite screen source, its clock and release law, and a compatible transport calculation would jointly determine cosmological initial data.

1 Scope

This technical companion states the inflation-free branch against the compact, screen-microphysics, dark-sector, and finite-source cosmic microwave background (CMB) papers.

2 Finite quotient theorem scope

Finite quotient ensemble theorem. Fix a finite regulator r. The physical presentation space is Σ_r, the presentation redundancy groupoid is Γ_r, and the finite physical quotient is

$$Q_r = \Sigma_r/\Gamma_r, \qquad \pi_r : \Sigma_r \to Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map n_r is a normal-form map, not a probability law. Any physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

Observable algebras and reference states. In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\mathrm{phys}}$ with state

$$\omega_r(A) = \mathrm{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\mathrm{phys}}, \tau_{r,b}^0\right), \qquad \mathcal{A}_{r,b}^{\mathrm{phys}} = z_{r,b} B(\widetilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \to U(\widetilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(A)}{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\widetilde{\mathcal{H}}_r \cong \bigoplus_\alpha V_\alpha \otimes M_\alpha,$$

with $d_\alpha = \dim V_\alpha$ and $m_\alpha = \dim M_\alpha$, then

$$\mathcal{A}_{r,b}^{\mathrm{phys}} \cong \bigoplus_\alpha I_{V_\alpha} \otimes B(M_\alpha), \qquad p_{r,\alpha} = \frac{d_\alpha m_\alpha}{\sum_\beta d_\beta m_\beta}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

OPH quotient ensemble. An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q)e^{-S_r(q)}, \qquad Z_r = \sum_{q\in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

Equivalently, one may state an intrinsic projective prior ν_r on Q_r and set $\mu_r = (n_r)_\#\nu_r$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

Normal-form projector non-selection. For any retraction $N : Q \to Q_{\mathrm{nf}}$ onto a subset $Q_{\mathrm{nf}} \subseteq Q$, that is, any map whose restriction to Q_{nf} is the identity, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_\#\mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on Q_{nf} is fixed; both statements use the retraction property. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

Selection-gap corollary. Let $X \subseteq Q_{\mathrm{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine X and its quotient-visible invariants, but they do not choose a member of X. If two laws μ, ν are supported on X and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

Finite MaxEnt quotient ensemble. For finite Q, positive m, and quotient observables $F_1, \ldots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state σ_r,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

Refinement compatibility and RG closure. For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse map. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' : c_{sr}(q') = q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q. Then

$$(c_{sr})_{\#} \mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#} \mu_{k+1} - \mu_k\|_{\mathrm{TV}},$$

then

$$\|(c_{nr})_{\#} \mu_n - \mu_r\|_{\mathrm{TV}} \leq \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} [\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$. If the residual is uniformly bounded by ε, the induced total-variation defect is bounded by $\tanh \varepsilon$.

Implementation invariance and representative lifting. If implementations A, B have quotient bijections $h_r : Q_r^A \to Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_\# \mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\mathrm{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x)\kappa_r(x \mid \pi_r x), \qquad \sum_{x:\pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_\# \widetilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

Quotient-lumpable kernels and sampler correctness. A representative kernel $\widetilde{P}(x, y)$ descends to Q_r only when

$$P_Q(q, q') = \sum_{y:\pi(y)=q'} \widetilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q)e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min\left\{1, \frac{w(q')R(q', q)}{w(q)R(q, q')}\right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

Repair generators are not selectors. A repair generator of the form

$$L_{\mathrm{rep}} = \sum_C c_C(I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations E_C are defined on $L^2(X_r, \pi_r)$, so the reference law π_r is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a positive lower bound γ_*, then

$$L_{\mathrm{rep}} \geq \gamma_* \kappa_r (I - P_0).$$

Finite repair completeness gives $\kappa_r > 0$ at fixed regulator. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt.

Finite evidence accuracy. For bounded coarse observables O, if

$$\|\widehat{\mu}_s - \mu_s\|_{\mathrm{TV}} \le \epsilon_{\mathrm{samp}}$$

and the refinement defects sum to ϵ_{ref}, then

$$\Big|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\Big| \le 2\|O\|_\infty(\epsilon_{\mathrm{samp}} + \epsilon_{\mathrm{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

Physical-vacuum requirement. A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. A physical-vacuum interpretation requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q, q') = J_r(q', q) \ge 0$, local potential V_r, and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q')(f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state Ω_r, and the finite vacuum law is

$$\mu_r^{\mathrm{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with μ_r^{vac}. A continuum interpretation additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family.

Primordial and cosmological prediction firewall. A screen covariance contains incomplete radial information. The complete one-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional kernel that persists under positivity. OPH physical primordial interpretation requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, physical mode, radial-null-space, and forward-projection receipts together with a scale-natural physical dilation intertwiner or complete radial cross-covariance tomography. A finite radial prior produces a conditional continuation. Observable CMB comparison also requires declared source, solver, dataset, covariance, nuisance, data-use, and pooled-reducer provenance.

Interpretation classes and required receipts. Every ensemble-facing run records its ensemble id, interpretation class, regulator, representative schema, gauge action, canonicalizer, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt rather than the ensemble definition. The interpretation classes are

seed, proposal, or repair noise,
conventional reference ensemble,
OPH-native quotient ensemble,
OPH physical vacuum,
OPH primordial field,
observable cosmological prediction.

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared.

3 Finite Screen Spectrum Theorem Package

The screen-level spectrum theorem supplies the finite source object used by the cosmology branch. It does not supply temperature-temperature, temperature-polarization, polarization-polarization, lensing, likelihoods, or a physical cosmic microwave background interpretation. Those require the Boltzmann and data-contract theorems.

Definition 3.1 (finite screen regulator)**.** *For regulator r, a finite screen regulator is*

$$\mathfrak{S}_r = (\mathcal{T}_r, M_r, \Gamma_r, \mathcal{Q}_r, C_{sr}, J_{rs}),$$

where $\mathcal{T}_r$ is a finite cellulation of the screen, $M_r \succ 0$ is the area/quadrature mass matrix, Γ_r is the hidden-presentation groupoid, $\mathcal{Q}_r = \Sigma_r/\Gamma_r$ is the physical quotient, and C_{sr}, J_{rs} are the coarse-graining and interpolation maps. Shape regularity and quadrature convergence require

$$f^{\mathsf{T}} M_r g \to \int_{S^2} f g \, d\Omega$$

with an explicit band-limited error bound

$$\left| f^{\mathsf{T}} M_r g - \int f g \, d\Omega \right| \le \varepsilon_{M,r}(L) \|f\|_{H^s} \|g\|_{H^s}.$$

Patch count alone is therefore not an angular-resolution certificate.

Definition 3.2 (geometric screen scalar)**.** *Let the quotient-visible collar-volume readout be*

$$J_{X,r}(x) = \lambda_r(x) \sqrt{\det \sigma_{AB,r}(x)} > 0,$$

and let $\bar{J}_{X,r}$ be emitted by an independently defined homogeneous or frozen-background branch, not by a CMB fit. Define

$$q_{0,r} = \frac{1}{3} \log \frac{J_{X,r}}{\bar{J}_{X,r}}.$$

On a certified spherical chart, with $B_r = [1, n_x, n_y, n_z]$,

$$\Pi_{<2,r} = B_r (B_r^{\mathsf{T}} M_r B_r)^{-1} B_r^{\mathsf{T}} M_r, \qquad q_r = (I - \Pi_{<2,r}) q_{0,r}.$$

On an irregular screen, B_r is replaced by the certified generalized eigenprojector onto the finite $\ell = 0, 1$ subspace. Hard-coded feature z-scores or observer-summary weights are diagnostic proxies, not q_r.

Proposition 3.3 (quotient invariance)**.** *If a recharting U preserves the screen inner product and geometric readout,*

$$U^\mathsf{T} M_r' U = M_r, \qquad q_{0,r}' = U q_{0,r}, \qquad B_r' = U B_r,$$

then $\Pi_{\geq 2,r}' U = U \Pi_{\geq 2,r}$, and $q_r' = U q_r$.

Proof. Substitution gives

$$\begin{aligned}\Pi_{<2,r}' U &= U B_r (B_r^\mathsf{T} U^\mathsf{T} M_r' U B_r)^{-1} B_r^\mathsf{T} U^\mathsf{T} M_r' U \\ &= U B_r (B_r^\mathsf{T} M_r B_r)^{-1} B_r^\mathsf{T} M_r = U \Pi_{<2,r}.\end{aligned}$$

Subtracting from U gives the high-mode identity. $\square$

Definition 3.4 (physical scalar precision and action)**.** *The scalar precision K_r must have a physical origin on $V_r = \operatorname{im} \Pi_{\geq 2,r}$. Allowed branches are:*

$$K_r^{\mathrm{Hess}} = \nabla^2 \Phi_r \Big|_{q=0},$$

for a quotient-visible mismatch or release free energy; a repair Dirichlet form

$$K_r^{\mathrm{diss}} = H_r - T_r^\mathsf{T} H_r T_r, \qquad T_r^\mathsf{T} H_r T_r \preceq H_r;$$

or the reversible generator form

$$K_r^{\mathrm{Dir}} = -\frac{1}{2}(L_r + L_r^{\dagger_H}).$$

A raw nonsymmetric repair matrix or caller-supplied κ is not a precision operator. The absolute normalization must be fixed independently, for example by

$$K_0 Y_{\ell m} = \ell(\ell+1) Y_{\ell m}$$

on the $\theta = 0$ branch. Once normalized,

$$S_{\mathrm{scr},r}[q] = \frac{1}{2A_{q,r}} \langle q, K_r q \rangle_r = \frac{1}{2A_{q,r}} q^\mathsf{T} M_r K_r q.$$

Theorem 3.5 (finite MaxEnt screen covariance)**.** *Let $K_r e_i = \kappa_i e_i$ with an M_r-orthonormal basis of V_r, $\kappa_i > 0$, and $d_r = \dim V_r$. Among continuous densities in coefficients $q = \sum_i a_i e_i$ with $\mathbb{E}[a_i] = 0$ and*

$$\frac{1}{2} \mathbb{E} \langle q, K_r q \rangle_r = E_{q,r}^{\mathrm{src}},$$

the unique entropy maximizer is

$$d\mu_r(q) = Z_r^{-1} \exp\left[-\frac{1}{2A_{q,r}} \langle q, K_r q \rangle_r \right] dq, \qquad A_{q,r} = \frac{2E_{q,r}^{\mathrm{src}}}{d_r},$$

and

$$\mathbb{E}[a_i a_j] = \delta_{ij} \frac{A_{q,r}}{\kappa_i}.$$

Proof. The Euler–Lagrange equation for entropy with normalization and quadratic-energy constraints gives a Gaussian density with inverse temperature β_r. The expected energy is

$$E^{\mathrm{src}}_{q,r} = \frac{1}{2}\sum_i \kappa_i \frac{1}{\beta_r \kappa_i} = \frac{d_r}{2\beta_r}.$$

Thus $A_{q,r} = \beta_r^{-1} = 2E^{\mathrm{src}}_{q,r}/d_r$. Strict concavity of entropy gives uniqueness. □

Remark 3.6 (discrete quotient branch). For a discrete finite quotient with intrinsic base weight $m_r(q)$, the exact Gibbs law is

$$\mu_r(q) = Z_r^{-1} m_r(q) e^{-\beta_r E_r(q)}.$$

A Gaussian statement on that branch requires a separate Laplace or central-limit theorem with controlled higher-order terms.

Definition 3.7 (source release energy)**.** *Let m_r^0 be the tracially pointed base weight on the finite physical collar quotient. Let $F_{r,a}$ be a finite ledger of primitive collar observables: scalar occupancy, protected-center presence, released volume, conserved source charges, and the release clock. The ledger excludes screen energy, sky spectra, CMB residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. For source-emitted constraint values $c^{\mathrm{col}}_{r,a}$, the selected release law is*

$$\nu_r^{\mathrm{rel}}(x) = \frac{m_r^0(x)\exp[-\sum_a \lambda_{r,a} F_{r,a}(x)]}{Z_r}, \qquad \mathbb{E}_{\nu_r^{\mathrm{rel}}} F_{r,a} = c^{\mathrm{col}}_{r,a}.$$

Strict concavity of relative entropy gives uniqueness on full support. A boundary optimum uses the same form on its certified support. The constraint ledger, base weight, source DAG, multipliers, support, and refinement maps belong to the receipt.

For this independently selected quotient ensemble on released collar normal forms, define

$$E^{\mathrm{src}}_{q,r} = \frac{1}{2}\int \langle q_r(x), K_r q_r(x)\rangle_r \, d\nu_r^{\mathrm{rel}}(x).$$

The receipt must expose ν_r^{rel}, the base measure, K_r, d_r, $E^{\mathrm{src}}_{q,r}$, $A_{q,r}$, no-observation ancestry, and the same operator normalization used in the action. MaxEnt alone does not determine amplitude.

Proposition 3.8 (microscopic-law necessity)**.** *For fixed positive K_r, the Gaussian family $Z(A)^{-1}\exp[-\langle q, K_r q\rangle/(2A)]$ exists for every $A > 0$. The quadratic MaxEnt form therefore leaves one positive scale free. The source release law in Definition 3.7, or an equivalent source-only vacuum law, supplies the energy that fixes $A_{q,r} = 2E^{\mathrm{src}}_{q,r}/d_r$.*

Theorem 3.9 (rotational screen spectrum)**.** *If the continuum precision K_θ commutes with the $SO(3)$ action on scalar functions, then by Schur's lemma each $\mathcal{H}_\ell$ is an eigenspace. For the exact conformal-shell family,*

$$\Lambda_\ell(\theta) = \frac{\Gamma(\ell + 2 + \theta/2)}{\Gamma(\ell - \theta/2)}, \qquad \ell \geq 2.$$

Then $\Lambda_\ell(0) = \ell(\ell+1)$, and the MaxEnt covariance gives

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

Consequently $\mathcal{D}_\ell^q = \ell(\ell+1)C_\ell^q/(2\pi) \sim (A_q/2\pi)\ell^{-\theta}$.

Definition 3.10 (source-derived conformal precision). *Let $L_r \succeq 0$ be the normalized detailed-balanced scalar collar Dirichlet operator on V_r, with continuum limit $-\Delta_{S^2}$. Set*

$$B_r = (L_r + \tfrac{1}{4}I)^{1/2}, \qquad K_{\theta,r} = \frac{\Gamma(B_r + \frac{3}{2} + \frac{\theta}{2})}{\Gamma(B_r - \frac{1}{2} - \frac{\theta}{2})}.$$

For $-2 < \theta < 4$ on the retained $\ell \geq 2$ branch, this operator is positive. The gamma recurrence gives $K_{0,r} = L_r$ exactly. The finite receipt carries detailed balance, positivity, the $\ell(\ell+1)$ normalization, anisotropy splitting, and refinement residuals.

Theorem 3.11 (finite refinement error). *Suppose that for $2 \leq \ell \leq L$*

$$\left|\frac{\lambda_{\ell m,r}}{\Lambda_\ell(\theta)} - 1\right| \leq \varepsilon_{K,r}(L), \qquad \left|\frac{A_{q,r}}{A_q} - 1\right| \leq \varepsilon_{A,r}, \qquad \varepsilon_{K,r} < 1.$$

Then $C^q_{\ell m,r} = A_{q,r}/\lambda_{\ell m,r}$ satisfies

$$\left|\frac{C^q_{\ell m,r}}{A_q/\Lambda_\ell(\theta)} - 1\right| \leq \frac{\varepsilon_{A,r} + \varepsilon_{K,r}}{1 - \varepsilon_{K,r}}.$$

Proof. Write $A_{q,r} = A_q(1 + a_r)$ and $\lambda_{\ell m,r} = \Lambda_\ell(1 + b_{\ell m,r})$. Then

$$\frac{C^q_{\ell m,r}}{A_q/\Lambda_\ell} = \frac{1 + a_r}{1 + b_{\ell m,r}},$$

and the displayed bound follows from $|a_r| \leq \varepsilon_A$, $|b_{\ell m,r}| \leq \varepsilon_K$. □

Theorem 3.12 (refinement-semigroup tilt). *Let R_b be the scalar refinement map for scale ratio $b > 1$. If one isolated scalar covariance mode has positive eigenvalue $\lambda(b)$, $\lambda(b_1 b_2) = \lambda(b_1)\lambda(b_2)$, and λ is continuous, then there is a unique real θ with*

$$\lambda(b) = b^{-\theta}, \qquad \theta = -\frac{\log \lambda(b)}{\log b}.$$

Proof. Set $g(t) = -\log \lambda(e^t)$. The semigroup law gives $g(t+s) = g(t) + g(s)$. Continuity makes $g(t) = \theta t$, hence the result. □

Definition 3.13 (edge-center reserve generator receipt). *Write $s = \log b$ for logarithmic refinement thickness. Let $u_{\mathrm{full}}(s)$ be the scalar-conditioned covariance-survival cocycle across a full oriented collar, and let $u_q(s)$ be its source-facing half-collar restriction. The receipt requires*

$$u(s+t) = u(s)u(t), \qquad u(0) = 1,$$

strong continuity at zero, the source-derived full-collar density

$$-u'_{\mathrm{full}}(0) = \frac{P_\star}{24},$$

and the orientation-reversal coarea identity

$$-u'_q(0) = \frac{1}{2}[-u'_{\mathrm{full}}(0)] = \frac{P_\star}{48}.$$

These are infinitesimal generator statements. A one-step survival probability has exponent $-\log u_q(\log b)/\log b$.

Theorem 3.14 (edge-center tilt and repair-clock reconciliation)**.** *Under Definition 3.13,*

$$u_q(s) = e^{-\theta s}, \qquad \theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}.$$

In the coordinate $\theta = \kappa_{\mathrm{rep}}(P_\star - \varphi)$, the same branch has

$$\kappa_{\mathrm{rep}}^{\mathrm{edge}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

The value e is a separate diagnostic hypothesis unless an additional identity equates it with this source-derived coordinate.

Proof. For $g(s) = -\log u_q(s)$, the cocycle law gives the continuous Cauchy equation. Hence $g(s) = \theta s$. Differentiation at zero and the half-collar identity give $\theta = P_\star/48$. The formula for $\kappa_{\mathrm{rep}}^{\mathrm{edge}}$ is algebraic. □

Theorem 3.15 (homogeneous radial source family)**.** *Assume source-stress closure, a single clock, freezeout, multicenter consistency, and translation/rotation covariance. Let $C_\zeta = M_{\Delta_\zeta^2}$ on the common physical $d\ln k$ mode basis. If a scale-natural source embedding transports refinement to physical dilation and its finite operator residual converges to*

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta, \qquad (D_s f)(k, \hat{k}) = f(e^{-s}k, \hat{k}),$$

then every positive measurable representative satisfies

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}$$

almost everywhere. Continuity gives the pointwise identity. A single shell has an infinite-dimensional radial kernel, including positive ambiguities; complete radial cross-covariances give the independent spherical-Hankel tomography route.

Proof. Conjugation of the multiplication operator gives $\Delta_\zeta^2(e^s k) = e^{-\theta s}\Delta_\zeta^2(k)$ almost everywhere. In logarithmic coordinates, rational-translation invariance on a common full-measure set forces $e^{\theta t}\Delta_\zeta^2(e^t)$ to be constant almost everywhere. The one-shell and tomography statements follow from the correlation-restriction and spherical-Hankel theorems in the primordial bridge packet. □

Theorem 3.16 (thin-shell power-law lift)**.** *Assume $q(\hat{n}) = Z_\star \Pi_{\ell \geq 2} \zeta_\star(R_\star \hat{n})$ and*

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}.$$

Then

$$C_\ell^q = 4\pi Z_\star^2 \int d\ln k\, \Delta_\zeta^2(k) j_\ell^2(kR_\star),$$

and, for $-2 < \theta < 4$,

$$A_q = \pi^{3/2} Z_\star^2 A_\zeta (k_\star R_\star)^\theta \frac{\Gamma(1+\theta/2)}{\Gamma(3/2+\theta/2)}.$$

Thus

$$A_\zeta = \frac{A_q}{\pi^{3/2} Z_\star^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}.$$

Proposition 3.17 (finite-window bound)**.** *Let $\Psi_\ell(k) = \int dr\, W(r) j_\ell(kr)$, $W \geq 0$, $\int W dr = 1$, with mean $R_\star$ and variance σ_R^2. If*

$$\delta_\ell(k) = \frac{k^2\sigma_R^2}{2} \sup_{r\in\operatorname{supp} W} |j_\ell''(kr)|,$$

then

$$|C^q_{\ell,W} - C^q_{\ell,\mathrm{shell}}| \leq 4\pi Z_\star^2 \int d\ln k\, \Delta_\zeta^2(k) \left[2\delta_\ell(k) + \delta_\ell(k)^2\right].$$

Proposition 3.18 (radial non-identifiability)**.** *For a finite radial basis, $C = Ap$. If $A \in \mathbb{R}^{N_\ell\times N_k}$ has rank r, then $\dim\ker A = N_k - r$, and every $p+v$ with $v \in \ker A$ gives the same screen spectrum. A physical radial reconstruction therefore requires either a source theorem restricting p, or a declared prior with a published null basis, resolution kernels, positivity checks, and prior-sensitivity report.*

Theorem 3.19 (promotion contract for the thin-shell power-law pipeline)**.** *Under the declared thin-shell lift and power-law source family above, one hash-locked source DAG may emit*

$$(q_r, S_{\mathrm{scr},r}, \theta, A_{q,r}, C^q_{\ell,r}, A_{\zeta,r}, \Delta^2_{\zeta,r})$$

as a conditional primordial packet when the geometric-scalar, collar-law, release-energy, reserve-generator, conformal-precision, refinement, source-stress, single-clock, freezeout, adiabaticity, isocurvature, phase-coherence, thin-shell, radial-null, finite-window, and forward- residual receipts pass on that DAG. Any measurement, likelihood, fitted parameter, or measurement-calibrated ancestor fails the source packet. Physical temperature and polarization spectra require the independent Boltzmann, recombination, nuisance, covariance, and likelihood gates. This contract is specific to the displayed pipeline. A different source-only radial reconstruction or forward projection may use another fully typed contract, provided it certifies its own injectivity or null space, window and convergence control, forward residual, and no-target-leak provenance.

Target 3.20 (requirements for a source-derived screen spectrum)**.** *A concrete source-derived spectrum requires the following receipt families before primordial interpretation:*

1. *geometry, scalar quotient, low-mode projector, scalar precision, and operator normalization;*
2. *quotient ensemble selection, scalar release energy, and MaxEnt screen covariance;*
3. *the infinitesimal reserve generator, scalar refinement tilt, and angular screen spectrum with a finite error budget;*
4. *source-stress, clock, freezeout, radial-window, radial-null, forward-residual, and transfer firewall receipts.*

The theorem and algorithm packet is complete for the declared thin-shell power-law pipeline. No finite source DAG that passes this receipt set is supplied, and the list is not a no-go for a differently typed radial pipeline.

4 Branch Thesis

The branch claim is:

On a finite-source branch, OPH may replace the logical jobs of inflation by clock-slice spatial-holonomy selection, same-boundary or low-k repair coherence, a geometric screen spectrum, MaxEnt release, and ordinary Boltzmann transfer.

This claim lies outside the recovered Standard Model and general-relativistic core. It becomes predictive only when the source objects are fixed before observational comparison. Screen synchronization implements modular self-consistency, with internal time generated by the state–algebra pair rather than by an external clock.

Coherent-matter same-channel forcing supplies a local scalar-channel identity for nondegenerate OPH-coherent material sources under Scalar Edge-Center Exhaustion. It does not supply the finite covariant dark-stress parent needed for cosmological w_A, $c_{s,A}^2$, σ_A, Q_A^μ, recipient-stress, gauge-variable, cold-dark-matter (CDM) limit, and refinement-convergence receipts.

The repair-charge dark-sector construction supplies a source-only $\bar{\rho}_A(a)$ only when a finite source fixes the anomaly abundance. Without that source, the dark continuation is unspecified.

5 Physical Conditions

Physical question	Required result or premise
Finite early state	Quotient normal-form branch with declared base measure and implementation invariance.
Flatness	Zero clock-slice spatial Levi–Civita holonomy identifies $\kappa = 0$. Exact selection requires a derivation from the finite screen or an explicit curvature premise. Damping gives only an approximate $\lvert K\rvert$ or $\lvert\Omega_K\rvert$ bound.
Horizon coherence	Same-boundary unique-extension theorem or a uniform low-k repair gap.
Near scale invariance	Geometric screen scalar, normalized scalar precision, source release energy, infinitesimal reserve-generator tilt, and angular screen-spectrum receipts.
Amplitude	The conditional theorem fixes $A_q = 2E_q^{\mathrm{src}}/d$ and its source-family conversion to A_ζ. A finite collar release law and energy receipt must supply their values without CMB data.
Hot start	MaxEnt release into the realized radiation branch with a common release clock.
Expansion history	Conformal invariance of the layered order and the a^4 record-density weight of Section 7; the record-production profile is a supplied datum, with the fixed-capacity de Sitter profile $a = -1/(H\eta)$ as one inhabitant.
Acoustic transfer	Declared source functions and a Boltzmann transfer calculation.

6 Numerical Acceptance Conditions

The simulator source is `https://github.com/muellerberndt/oph-physics-sim`, and public data archives are indexed at `https://github.com/FloatingPragma/observer-patch-holography/tree/main/evidence`. For this paper the simulator may emit diagnostic screen spectra, observer-screen synchronization receipts, and release-amplitude candidates. It supports an inflation-replacement interpretation only when one finite source supplies the primordial growth mode, physical radial lift, transfer calculation, and likelihood inputs.

7 Expansion as Record Density

The spacetime construction supplies a flat causal-order and count-volume limit for the conservative golden-record population and the retrospective count clock. The following statements transfer that family to a spatially flat expanding background. They are exact and machine-checked in the Lean module named below; the profile law that selects how many records a comoving cell produces per conformal tick is a supplied datum, and no statement below derives the Friedmann equation.

7.1 Expansion as record density on the source order

The spatially flat Friedmann–Lemaître–Robertson–Walker metric in conformal time η is

$$ds^2 = \sigma(\eta)^2\big(c^2\, d\eta^2 - |dx|^2\big), \tag{1}$$

with scale factor $\sigma(\eta) > 0$; the letter a is the edge radius of the read law. This metric is a positive conformal factor times the flat metric in the coordinates (η, x). A positive conformal factor leaves the causal order of a Lorentzian metric unchanged, and the causal order fixes the metric up to such a factor (Hawking, King and McCarthy 1976; Malament 1977). On the spatially flat family the conformal factor is the entire background, so every profile σ has the causal order of flat space in comoving coordinates. The layered read law of the source-net construction places an event at every (j, s) for a site s in a population S, lets $(j + 1, t)$ read (j, s) whenever $\|t - s\| \le a$, and takes the reflexive transitive closure of the reads as the order. Written in comoving coordinates with the tick $\Delta = a/c$ in conformal time, this law generates one finite order for every profile. The profile enters only through the count measure.

Proposition 7.1 (Conformal invariance of the layered order)**.** *Let S be a subset of a real normed space, $a > 0$ an edge radius and $\sigma_j > 0$ the scale at layer j. The physical position of the site s at layer j is $\sigma_j s$ and the physical read radius at that layer is $\sigma_j a$. The physical read from (j, s) to $(j + 1, t)$,*

$$\|\sigma_{j+1}t - \sigma_{j+1}s\| \le \sigma_{j+1}a,$$

holds exactly when $\|t - s\| \le a$. The layered orders generated by the physical reads and by the comoving reads coincide, for every profile $(\sigma_j)_{j\ge0}$.

Proof. For $\sigma > 0$, $\|\sigma t - \sigma s\| = \sigma\|t - s\|$, and multiplication by σ preserves and reflects the comparison with σa. The two read relations are equal, hence so are their reflexive transitive closures. □

Proposition 7.2 (FLRW count measure)**.** *Assign a comoving cell of volume v_s to every site s and the conformal tick Δ to every layer. Under a constant physical event density ρ, the physical tick at layer j is $\sigma_j\Delta$ and the physical cell volume is $\sigma_j^3 v_s$, so a finite set I of events carries the mass*

$$M_\sigma(I) = \rho \sum_{(j,s)\in I} (\sigma_j\Delta)(\sigma_j^3 v_s) = \rho \sum_{(j,s)\in I} \sigma_j^4\, \Delta\, v_s, \tag{2}$$

the discrete form of the four-volume weight $\sigma^4\, d\eta\, d^3x$. Write $M_1(I) = \rho\Delta \sum_{(j,s)\in I} v_s$ for the flat mass. If $0 \le \sigma_{\min} \le \sigma_j \le \sigma_{\max}$ on the layers of I, then

$$\sigma_{\min}^4 M_1(I) \le M_\sigma(I) \le \sigma_{\max}^4 M_1(I). \tag{3}$$

A constant profile $\sigma_j = \sigma$ gives $M_\sigma(I) = \sigma^4 M_1(I)$.

Proof. The identity in (2) is the product of the two scaled factors. Fourth powers are monotone on nonnegative reals and $\Delta v_s \geq 0$, so each term of the sum lies between the same term with σ_j replaced by $\sigma_{\min}$ and by $\sigma_{\max}$. Summing proves (3); the constant case is the common factor σ^4. $\square$

Corollary 7.3 (Count clock in an expanding family)**.** *Let I and J be finite event sets with uniform cells, J nonempty, with $\sigma^I_{\min} \leq \sigma_j \leq \sigma^I_{\max}$ on the layers of I, $\sigma^J_{\min} \leq \sigma_j \leq \sigma^J_{\max}$ on the layers of J, $\sigma^I_{\min} \geq 0$ and $\sigma^J_{\min} > 0$. Then*

$$\frac{\sigma^I_{\min}}{\sigma^J_{\max}} \left(\frac{|I|}{|J|}\right)^{1/4} \leq \left(\frac{M_\sigma(I)}{M_\sigma(J)}\right)^{1/4} \leq \frac{\sigma^I_{\max}}{\sigma^J_{\min}} \left(\frac{|I|}{|J|}\right)^{1/4}. \tag{4}$$

Let I and J approximate comoving diamonds of conformal durations T_I and T_J in the sense of the count-clock theorem for the flat order, so that $(|I|/|J|)^{1/4} \to T_I/T_J$ under refinement. A comoving worldline through a diamond of conformal duration T accrues proper time between $\sigma_{\min}T$ and $\sigma_{\max}T$. In the refinement limit the clock read from the physical masses, $(M_\sigma(I)/M_\sigma(J))^{1/4}$, and the proper-time ratio τ_I/τ_J lie in one interval whose endpoints have the quotient $(\sigma^I_{\max}/\sigma^I_{\min})(\sigma^J_{\max}/\sigma^J_{\min})$; the two readings agree up to this factor in either direction. For a profile continuous in η and diamonds shrinking to points, the factor tends to one and the clock reads the proper-time ratio exactly in the limit.

Proof. Divide the two sandwich inequalities (3) for I and J, with $M_1(I)/M_1(J) = |I|/|J|$ for uniform cells, and take the increasing fourth root; the fourth root of σ^4 is σ. The proper time along the comoving worldline is $\int \sigma \, d\eta$ over the conformal duration, bounded by the extreme scales times T; the ratio of two such enclosures has the stated endpoint quotient. Continuity of σ forces $\sigma_{\max}/\sigma_{\min} \to 1$ on shrinking diamonds. $\square$

Corollary 7.4 (Redshift as a count ratio)**.** *Let two congruent comoving diamonds, with the same comoving cells and the same conformal ticks, lie at the emission epoch, of scale σ_e, and at the reception epoch, of scale σ_0, the profile being constant across each. Their masses are $n_e = \kappa\sigma_e^4$ and $n_0 = \kappa\sigma_0^4$ with the common factor $\kappa = \rho\Delta \sum_{(j,s)\in I} v_s$, and*

$$1 + z = \frac{\sigma_0}{\sigma_e} = \left(\frac{n_0}{n_e}\right)^{1/4}. \tag{5}$$

Proof. The constant case of Proposition 7.2 gives both masses; κ cancels in the ratio, and the positive fourth root of $(\sigma_0/\sigma_e)^4$ is σ_0/σ_e. $\square$

Remark 7.5 (Expansion history as a record-production law). On this family the Friedmann equation is a statement about record production: a comoving cell of volume v produces $\rho\,\sigma(\eta)^4\,\Delta\, v$ records per conformal tick, and a profile $\sigma(\eta)$ is the same datum as a record-production rate per comoving cell. The de Sitter profile $\sigma(\eta) = -1/(H\eta)$ on $\eta < 0$ solves $d\sigma/d\eta = H\sigma^2$, and its record density per comoving cell is $n(\eta) \propto (H\eta)^{-4}$; the fixed-capacity branch of the corpus, $w = -1$, corresponds to this profile. The Lean module `Lean/Geometry/SourceNetConformalRecordDensity.lean` checks the order invariance, the mass identity and sandwich, the clock enclosure (4), the redshift identity and the de Sitter derivative and density identities over the reals.

Three items lie outside this construction. The source law selecting the profile, the record-production rate per comoving cell, is not derived; the profile is a supplied datum. A reweighted flat order determines at most the conformal factor; on the spatially flat family that factor is the entire background, and spatial curvature $k \neq 0$ changes the comoving spatial metric, which this construction does not treat. No physical clock and no population selection by native repair follow from these identities.

8 Flatness Boundary

The flatness branch is split into identification and selection. On a clocked Friedmann–Lemaître–Robertson–Walker (FLRW) continuation with $u_a = -N\nabla_a\chi$, $h_{ab} = g_{ab}+u_a u_b$, and $K(\tau) = \kappa/a(\tau)^2$, vanishing area-normalized small-loop holonomy of the spatial Levi–Civita connection identifies $K = 0$ and hence $\kappa = 0$. It does not select that sector from the raw FLRW fiber. A numerical calculation must distinguish exact selection derived from the finite screen, a flat branch conditioned on an explicit premise, and a branch for which no selection theorem is supplied. Screen and collar defect decay is a repair diagnostic rather than a spatial curvature holonomy. Curvature damping supplies only an approximate bound on $|K|$ or $|\Omega_K|$.

9 Screen-Spectrum Branch

The load-bearing screen result is the angular screen covariance. It is distinct from a CMB spectrum:

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

This statement requires the geometric q_r of Definition 3.2, a normalized conformal precision operator $K_{\theta,r}$, the primitive collar release law, $E^{\mathrm{src}}_{q,r}$, the MaxEnt covariance theorem, and a finite-to-continuum refinement error budget. The edge-center theorem gives

$$\theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}, \qquad \kappa^{\mathrm{edge}}_{\mathrm{rep}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

Its premise is a source-emitted full-collar generator density $P_\star/24$ with the orientation-half identity. A finite one-step survival probability is converted through $-\log u/\log b$. The $e(P_\star - \varphi)$ clock branch is a diagnostic alternative. The finite generator and clock receipt is not supplied here.

At the comparison pixel $P_\star = 1.6309682094$, located from the measured fine-structure constant, the edge-center value is $n_s = 0.96602$; the forward pixel 1.6309721 moves it by less than 10^{-7}. The retrospective comparison with public temperature and polarization data is recorded in the cosmology postdiction ledger released with these papers: the value lies 0.27σ above Planck 2018, 0.69σ above SPT-3G with Planck, 0.79σ below the SPT, Planck and ACT combination, 1.28σ below ACT DR6 with Planck, and 2.4σ below ACT DR6 with Planck, lensing and baryon acoustic oscillations. The diagnostic $e(P_\star - \varphi)$ branch gives 0.96484. These are seen-data comparisons of a receipt-conditional value and select neither branch; a prospective test needs the frozen combination, pivot and thresholds of the primordial registration proposal released with the ledger. On the same branch the dilation cocycle gives zero running, the rank-one single-clock normal form gives zero source-side isocurvature, and the scalar screen field emits no transverse-traceless source, so the primordial tensor amplitude is zero. Planck 2018 gives a running of -0.0045 ± 0.0067 and ACT DR6 with Planck, lensing and baryon acoustic oscillations 0.0062 ± 0.0052; BICEP/Keck 2018 bounds $r_{0.05} < 0.036$ and the fourth Planck release with BK18 and BAO $r < 0.032$; Planck 2018 bounds the uncorrelated cold-dark-matter isocurvature fraction at 3.5 to 3.9 percent. The tensor zero and the running zero are the two rows on which the branch differs from single-field slow-roll inflation, and the tensor row is the one a null at the LiteBIRD design sensitivity would discriminate. The exponential form of the survival cocycle and the tilt $P_\star/48$ from the half-collar density, together with the exact rational brackets of every comparison above, are machine-checked in the Lean modules `EdgeCenterTiltCocycle` and `CosmologyLedgerBrackets`.

Under the declared source dilation cocycle, the primordial power law

$$\Delta^2_\zeta(k) = A_\zeta \left(\frac{k}{k_\star}\right)^{-\theta}$$

is the unique member of the one-dimensional source family. The exact thin-shell identity fixes the amplitude conversion

$$A_\zeta = \frac{A_q}{\pi^{3/2}} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}$$

at the source pivot when $q = \Pi_{\geq 2}\zeta(R_\star \hat{n})$ and $k_\star R_\star = 1$. For the general attachment $q = Z_\star \Pi_{\geq 2}\zeta(R_\star \hat{n})$, the displayed expression is divided by $Z_\star^2(k_\star R_\star)^\theta$, with $Z_\star \neq 0$ and $-2 < \theta < 4$. A finite run requires $R_\star$, the radial window, the unrestricted null-space report, the forward residual, and the physical scale. The screen theorem proves no temperature, polarization, lensing, acoustic-peak, or likelihood statement.

10 Unsupplied Premises

The same-boundary or low-k coherence branch has no quotient-language derivation. A finite source emitting the geometric collar-volume field

$$q_r = \Pi_{\geq 2,r}(1/3)\log(J_{X,r}/\bar{J}_{X,r})$$

is not constructed by the stated source data. The same source must supply the primitive collar release law and pooled release energy that fix A_q, together with the full-collar infinitesimal reserve generator and orientation-half condition used by the conditional $P_\star/48$ theorem. Its source-dependency graph must contain compatible stress, clock, freezeout, physical-scale, radial-window, null-space, and forward-residual evidence. This scientific prerequisite graph is not the semantic read-from order of the consensus layer. A mapping from its evidence objects to semantic commits, together with complete certified read supports, would let the generated informational order verify execution dependencies; it would not derive the evidence objects or their scientific relations. The resulting quantities are the inputs to the cosmic microwave background transfer and likelihood calculations. The source Gram space supplies a Euclidean spatial carrier of dimension three. After adjoining a real axis, a nonzero closed convex pointed influence cone invariant under the proper icosahedral rotations and all rapidities of one boost axis is the future or past Lorentz cone. Time orientation selects the future cone. The invariance concerns operational transformations of histories, interventions and clock readouts, beyond an algebraic action on source frames [1].

A conservative Fibonacci-record family has flat $1+3$ causal-order and normalized count-volume limits under a specified population, complete local neighbour reads and layer duration. Given complete authenticated ancestry, interval anchors and a nondegenerate reference interval in the same history, $(|I|/|J|)^{1/4}$ converges to the proper-time ratio of interior timelike diamonds. This retrospective clock uses order and cardinality, without coordinates or timestamps in its definition. The reference interval fixes relative units, while the population and read law remain assumptions of the continuum construction [1].

An inflationary realization requires the same events and clock to support physical expansion, stress, freezeout and scale, together with the radial, transfer and likelihood conditions above. The finite tensor theorem obtains an Einstein-shaped identity from nine supplied null balances, symmetry, discrete conservation and connectedness in a separate $3+1$ tensor interface. It does not identify tensor-coordinate differences with source event displacements. Smooth Einstein promotion requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification; scalar-curvature convergence alone is diagnostic. Neither cone covariance nor the flat count-clock law supplies those inflationary attachments.

References

[1] B. Müller et al., *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.tex.

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